

Reviewer-2 audit — Chapter 1 (current state, 2026-07)

Fresh cold read of Chapitrel1/redaction/manuscript_chap1_EN.md and its pipeline, after this session's changes (sensitivity not Shapley; v3.2.3 iter + station forcing; Fig 1↔2 swap; appendices A–G with new B = ERA5 bias and G = wind; Appendix D switched gap-fraction → fCover). Supersedes most of review/pipeline_critique.md (that was the Shapley / r=0.93 era).

Tiers: **T1 science at risk** · **T2 missing analysis** · **T3 presentation/provenance**. Each item: severity · evidence · fix. **[NEW]** = introduced or changed this session.

T1 — Science at risk

1.1 [HIGH] [NEW] The open-canopy failure is described with the wrong sign

The manuscript repeatedly says the model “reverses sign in the sparsest stands, holding the understory near neutral **where the loggers cool**” (§3.2, §3.3, §4.2, Abstract), and Appendix D now says the model “**under-cools** the open, low-cover stands.”

Evidence (data). For the low-cover plots that dominate the residual (fCover < 0.6): obs $\Delta T_{\max}$ mean = **+3.81 °C** (strong **warming / amplification**), model = **+0.22 °C** (neutral); per plot 41_18 obs +4.80/sim +0.21, 41_39 +4.53/+0.23, 41_49 +4.22/+0.18. High-cover plots: obs −1.44, model −0.25. So the open failure is the model **failing to reproduce the observed warming (amplification)**, not “loggers cool / model reverses sign,” and it is **not** “under-cooling” (the model is *cooler* than obs there, not warmer).

Why it matters. This is the headline description of the glass ceiling; it is currently backwards for exactly the plots invoked. The floor mechanism (below) is coherent with the *correct* direction, so only the prose needs fixing — but it must be fixed everywhere (Abstract, §3.2/3.3, §4.2, App D).

Fix. Reword to: “in the open, low-cover stands the loggers **amplify** (understory hotter than the open), while the model holds them near neutral — it fails to reproduce the amplification.” Replace “under-cools the open stands” with “fails to warm / under-amplifies.”

1.2 [HIGH] [NEW] The fCover-floor residual (Appendix D) is a weaker, partly self-inflicted argument than the gap fraction it replaced

Switching Fig D1 from an external gap-fraction metric to the model's **own fractional-cover input, floored at 0.5**, regresses the residual on a *model input* mixed with external covariates. Part of the correlation is mechanical: the 4 open plots are pinned at the floor and cannot respond by construction, so “residual is largest at the floor” is partly “we clipped our own input.” The old gap-fraction version regressed on something the model genuinely **cannot see** — a cleaner “1-D limitation” claim.

Why it matters. It invites the obvious reviewer move: “then lower the floor and re-run the open plots.” It is the single sharpest actionable thread in the paper and it connects straight to 1.1.

Fix (choose one). - (a) Revert Appendix D to the external openness metric (gap fraction, or raw uncapped cover $\approx 1 - \text{gap}$), keeping the honest “structure the model can’t see” framing; or - (b) Keep fCover but **test lowering the floor**: re-run the 4 floored open plots at their real cover and report whether the amplification failure shrinks. If it does, that is a real, fixable model limitation (strong result); if it does not, the floor is not the cause and the wording in 1.1 must drop the floor mechanism. Decide before submission; do not ship the current half-way version.

1.3 [HIGH] The “uniform offset cancels, so the ranking survives” defence is in tension with a metric-dependent model

The model captures $\sim 13\%$ of the amplitude and reverses behaviour by regime, yet §4.2 defends the ranking with “a fixed offset cancels in the up-minus-down difference.” A **sign reversal in the open regime** and a model that gets the **slope right ($0.90 \approx 0.90$, $r=0.78$) but the amplitude wrong (13 %)** are, by definition, **non-uniform / metric-dependent** compression — exactly what the offset argument does not cover. The manuscript half-admits this (the “trait-dependent compression” caveat) but leaves the tension unresolved.

Mitigation that already exists — state it. The ranking **is** shown in the faithful (closed) regime: Fig 3 P4 has LAI $-0.28^\circ\text{C}/\text{SD}$ vs profile ≈ 0 . So “LAI $\square$ profile” holds on the plots the model gets right, independently of the open-canopy failure. Say this explicitly as the rebuttal, and lean the ranking on the two **model-free** lines (Fig 5 forward, Fig 6 nested) rather than on MuSICA’s radiative gain.

Fix. Reconcile “slope-right/amplitude-wrong” with “compression $\approx$ uniform” in one sentence; state the closed-canopy-only ranking as the robustness check; make explicit that the ranking’s load is borne by the model-free evidence, not the under-buffering model.

T2 — Missing analyses

2.1 [MEDIUM] No uncertainty on the central result (Fig 3)

The attribution violins are distributions, but there is **no CI/bootstrap** on the per-archetype median sensitivity per trait, and no test that LAI > profile is significant. “LAI dominates, profile weakest” rests on point estimates — a standard reviewer ask. **Fix.** Bootstrap plots within each archetype, report a CI on each trait’s median sensitivity, and confirm the LAI interval does not overlap the profile interval (per archetype). One extra panel or a table.

2.2 [MEDIUM] Forward inclusion (Fig 5) is order-dependent

Fixed order [LAI, $H_{\max}$, fCover, LAD] guarantees LAD-adds-least. The **order-free** answer is Fig 6 (nested regression: structure adds $\Delta R^2 = 0.00$). Right now Fig 5 carries an order-dependent claim and Fig 6 rescues it only implicitly. **Fix.** One sentence: name Fig 6 as the order-free confirmation of the forward-inclusion order; optionally show LAD-first as a robustness inset.

2.3 [MEDIUM] Per-archetype forward r on $n \approx 8$ is near-noise

Fig 5's per-cluster lines and the “ -0.09 to $+0.22$ ” per-archetype LAD increments sit on $n = 8-15$. Already caveated as “small per-archetype subsets,” but the coloured per-cluster lines invite over-reading. **Fix.** Annotate n per cluster, or grey the per-cluster lines and keep the pooled “All” as the claim.

2.4 [LOW] H2 controlled test (Fig C1) is at 4 centroids, not per plot

Everywhere else the design is per-plot to avoid Jensen bias ($f(\text{mean}) \neq \text{mean}(f)$); the H2 test is at the 4 archetype centroids. Defensible (it is a counterfactual sweep), but the asymmetry should be named. **Fix.** One clause acknowledging Fig C1 is centroid-based.

2.5 [LOW] Night-time buffering ($\Delta T_{\min}$) not examined

Longwave trapping at night may weight height/profile differently than daytime shading; refugia care about both extremes. Acknowledged as a limitation — keep, but consider flagging that the profile could matter more at night (an honest bound on the “profile is negligible” headline).

T3 — Presentation / provenance

3.1 [CLEARED, keep visible] Provenance is now consistent

Attribution (metrics6_frblo), validation and forward (tab_hobo_frblo_validation.csv, musica_hobo_z05_v323station_iter, forcing FR-Blo_2021_v2.nc) all use the **same station forcing and v3.2.3**. The prior critique's “three simulation branches” concern is resolved. The one **intentional** difference is LAD resolution — 1 m for the attribution, 0.5 m for the validation/robustness — which §4.4 justifies as conservative. Keep that sentence; it is the one provenance seam a reviewer could pull.

3.2 [LOW] Number-consistency after the session's edits — OK, but re-sweep once

fCover $r = -0.80$ / adj $R^2 = 0.71$, wind $z0 = 0.44$ / factor ~ 0.4 / $\sim 2.3\times$, Moran 0.07 (resid) / 0.15 (field) are consistent across Abstract, §3.3, §4.2, App D, App G. The Abstract now reflects the fCover/wind edits. Do a final numeric grep before submission.

3.3 [LOW] Internal filename/label crossing

Figure 1 → Fig2_site_typology_composite.png, Figure 2 → Fig1_methodo.png, and the appendix files are letter-shifted (Fig C1 → B1_h2_...png, etc.). Invisible to readers, but a maintenance hazard. **Fix (optional)**. Rename files to match labels in one pass, or add a mapping note in the repo.

Discussion points worth adding (the user's "pts de disc")

- **Open stands as heat traps, not refugia.** The amplification the model misses (1.1) is ecologically meaningful: open/gappy stands can be *hotter* than the open field. The management message ("keep leaf area and closure") is reinforced by this, and the model's blindness to it is a stated bound on using the model for management — currently framed only as error, could be framed ecologically.
- **Species/leaf-habit scope.** The "where foliage sits is redundant once quantity is known" result is for a *closed deciduous oak* canopy. Evergreen/conifer stands with deeper, more differentiated crowns are exactly where the vertical profile might still matter — a one-line honest hedge strengthens the claim rather than weakens it.
- **The cover floor as a concrete model-improvement target** (ties to 1.2): naming it turns a weakness into a perspective ("relaxing the sub-canopy cover floor is the first step to closing the open-canopy gap").

Priority order before the jury / submission

1. **1.1** fix the open-canopy sign wording everywhere (quick, and currently wrong).
2. **1.2** decide gap-fraction-revert vs fCover-with-floor-test for Appendix D (the sharpest thread).
3. **1.3** reconcile slope-right/amplitude-wrong vs "uniform offset"; state the closed-canopy ranking.
4. **2.1** add a CI/bootstrap to Fig 3.
5. **2.2/2.3** name Fig 6 as the order-free answer; tame the $n \approx 8$ per-cluster lines.